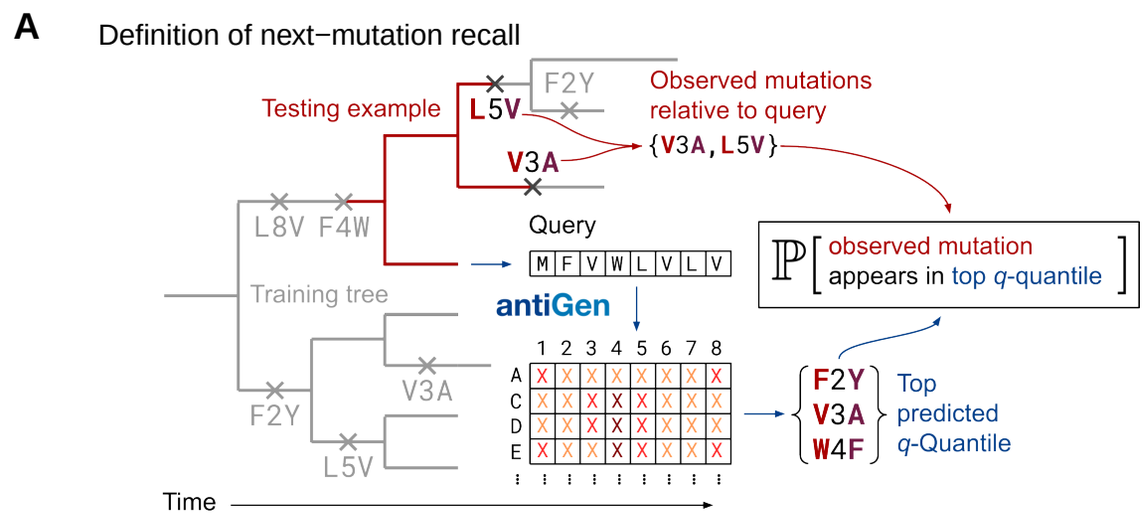
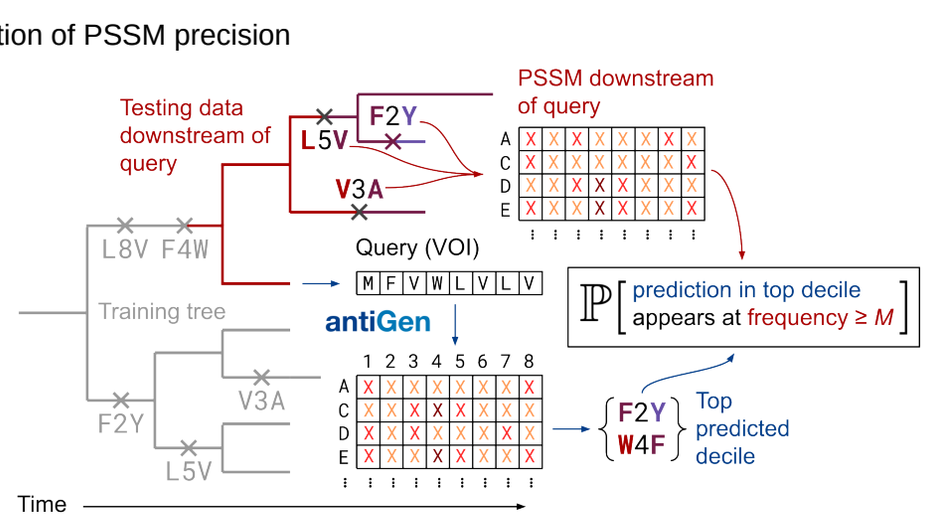
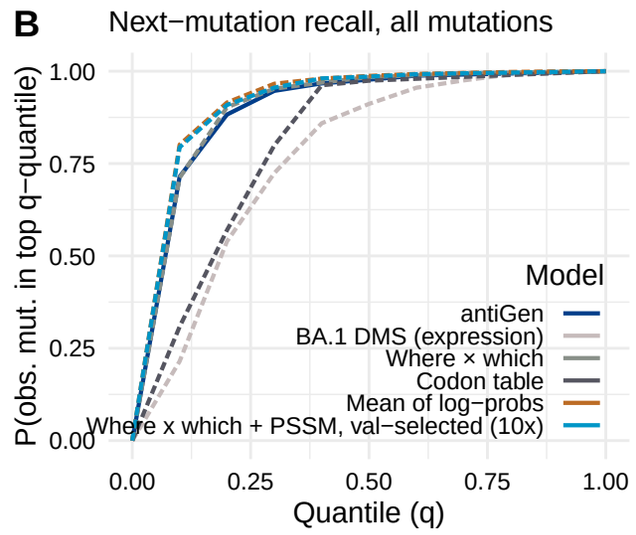
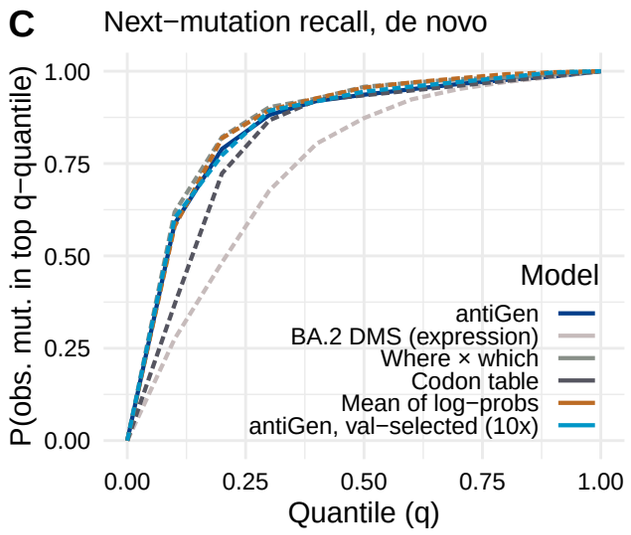
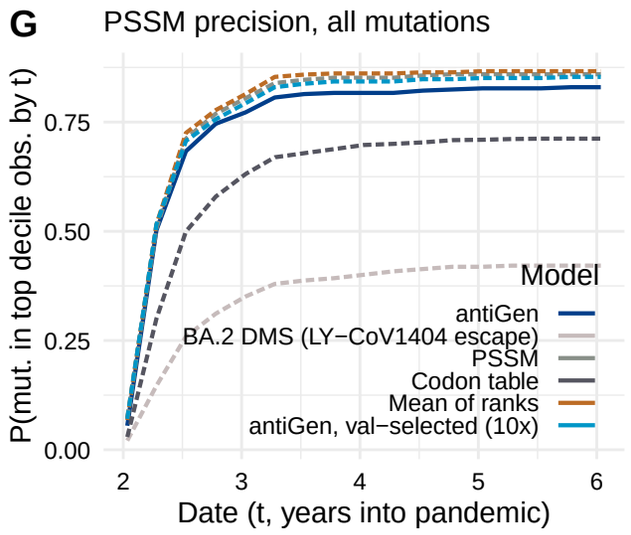
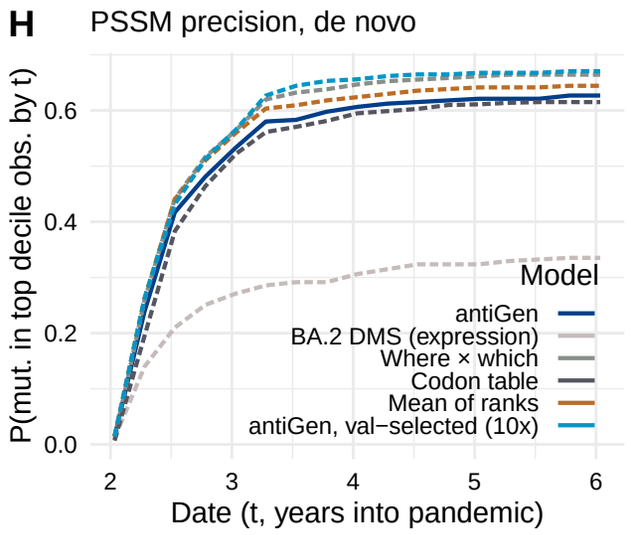

A
Definition of next−mutation recall
Testing example
L5V
F2Y
V3A
Observed mutations relative to query
{V3A, L5V}
L8V F4W
Query
M F V W L V L V
Training tree
antiGen
1 2 3 4 5 6 7 8
A
C
D
E
ℙ[observed mutation appears in top q-quantile]
F2Y
V3A
W4F
Top predicted q-Quantile
Time

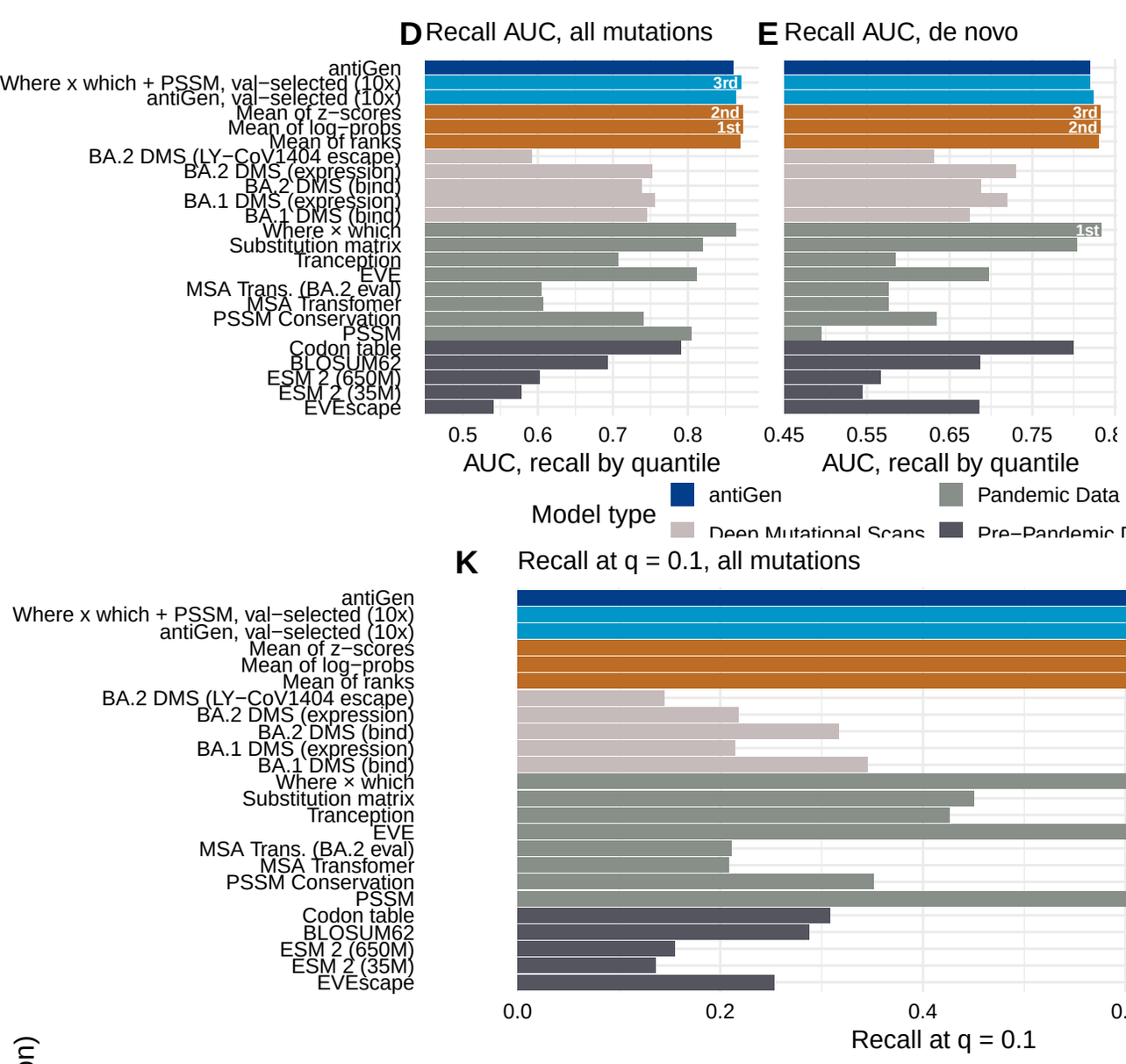

F
Definition of PSSM precision
Testing data downstream of query
PSSM downstream of query
L5V
F2Y
V3A
L8V F4W
Query (VOI)
M F V W L V L V
Training tree
antiGen
1 2 3 4 5 6 7 8
A
C
D
E
ℙ[prediction in top decile appears at frequency ≥ M]
F2Y
W4F
Top predicted decile
Time


B
Next−mutation recall, all mutations
P(obs. mut. in top q−quantile)
1.00
0.75
0.50
0.25
0.00
Model
antiGen
BA.1 DMS (expression)
Where × which
Codon table
Mean of log−probs
Where x which + PSSM, val−selected (10x)
0.00 0.25 0.50 0.75 1.00
Quantile (q)

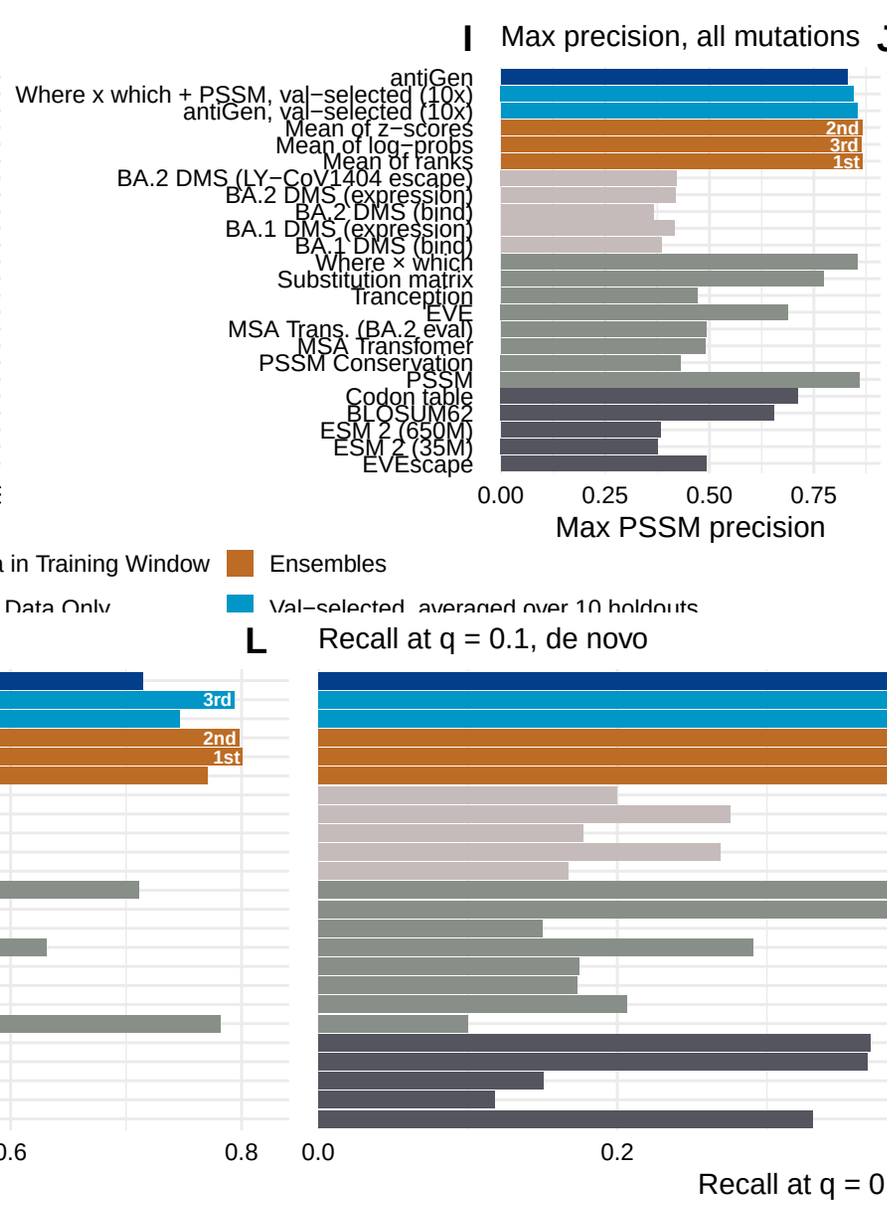

C
Next−mutation recall, de novo
P(obs. mut. in top q−quantile)
1.00
0.75
0.50
0.25
0.00
Model
antiGen
BA.2 DMS (expression)
Where × which
Codon table
Mean of log−probs
antiGen, val−selected (10x)
0.00 0.25 0.50 0.75 1.00
Quantile (q)

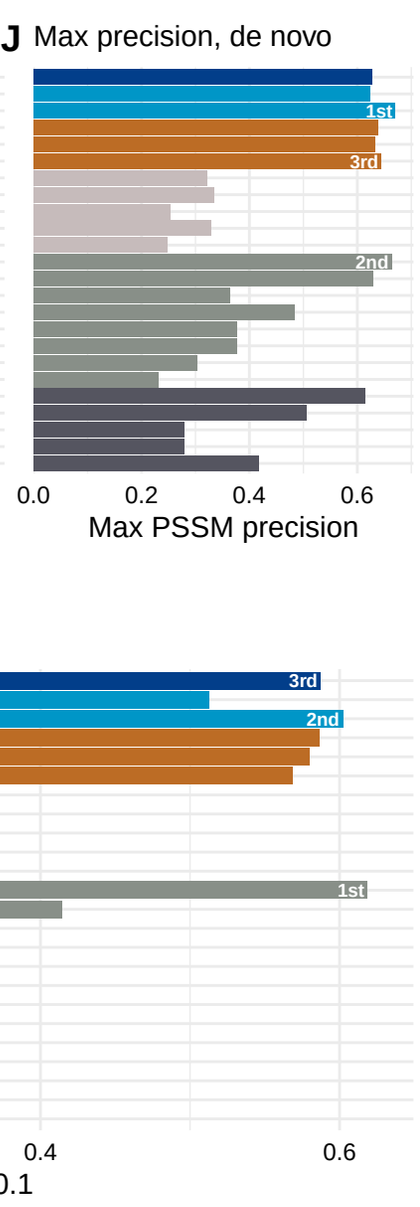

G
PSSM precision, all mutations
P(mut. in top decile obs. by t)
0.75
0.50
0.25
0.00
Model
antiGen
BA.2 DMS (LY−CoV1404 escape)
PSSM
Codon table
Mean of ranks
antiGen, val−selected (10x)
2 3 4 5 6
Date (t, years into pandemic)


H
PSSM precision, de novo
P(mut. in top decile obs. by t)
0.6
0.4
0.2
0.0
Model
antiGen
BA.2 DMS (expression)
Where × which
Codon table
Mean of ranks
antiGen, val−selected (10x)
2 3 4 5 6
Date (t, years into pandemic)

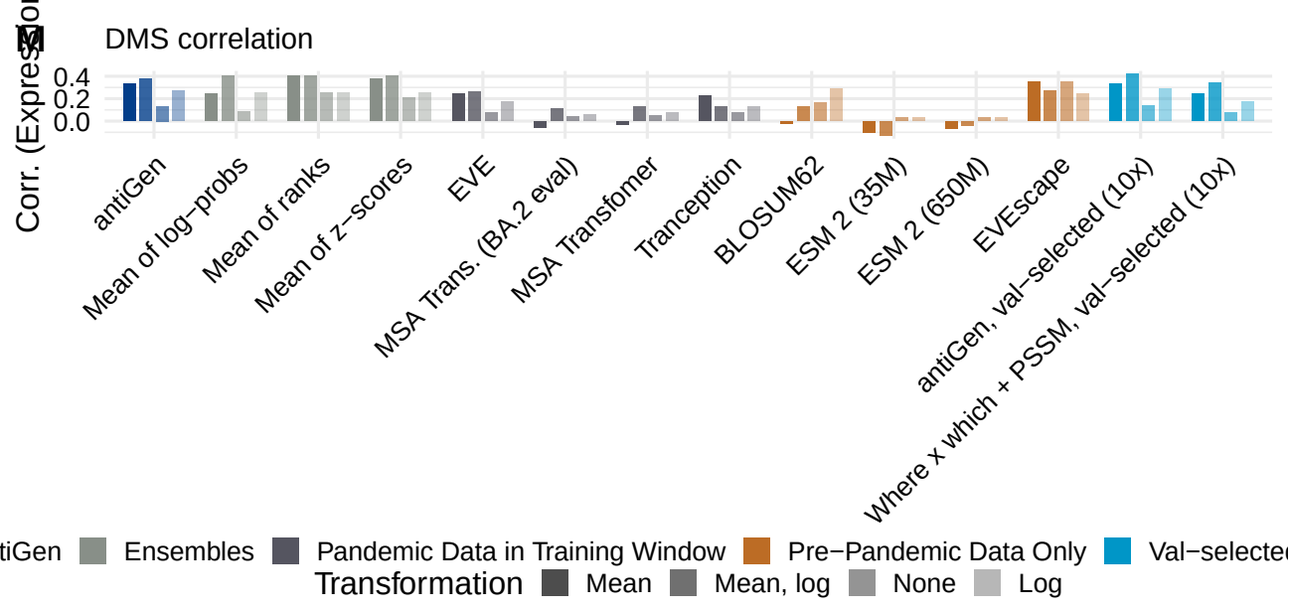

D
Recall AUC, all mutations
antiGen
Where x which + PSSM, val−selected (10x)
antiGen, val−selected (10x)
Mean of z−scores
Mean of log−probs
Mean of ranks
BA.2 DMS (LY−CoV1404 escape)
BA.2 DMS (expression)
BA.2 DMS (bind)
BA.1 DMS (expression)
BA.1 DMS (bind)
Where × which
Substitution matrix
Tranception
EVE
MSA Trans. (BA.2 eval)
MSA Transfomer
PSSM Conservation
PSSM
Codon table
BLOSUM62
ESM 2 (650M)
ESM 2 (35M)
EVEscape
3rd
2nd
1st
0.5 0.6 0.7 0.8
AUC, recall by quantile

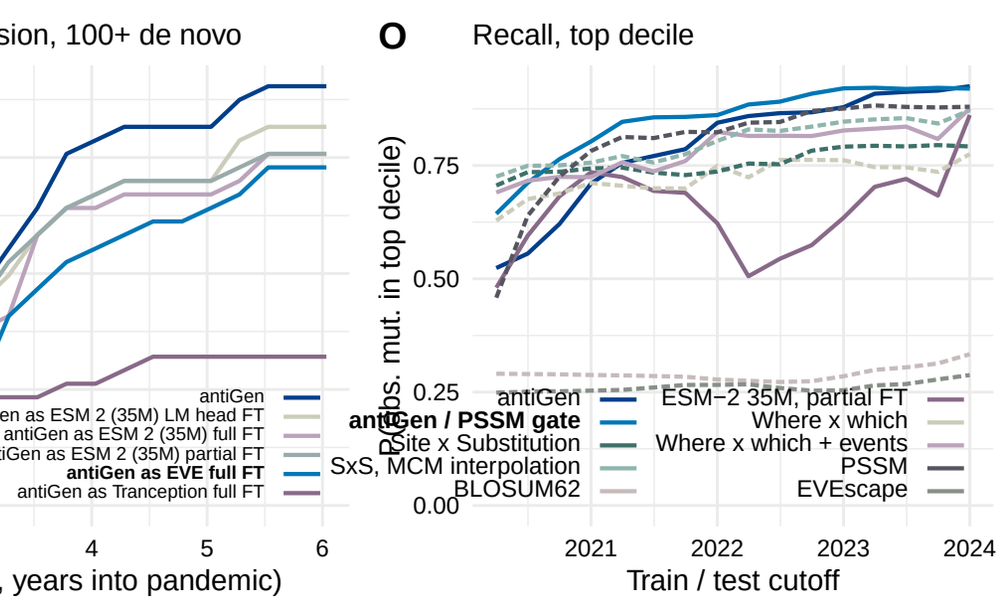
E
Recall AUC, de novo
3rd
2nd
1st
0.45 0.55 0.65 0.75 0.8
AUC, recall by quantile


I
Max precision, all mutations
antiGen
Where x which + PSSM, val−selected (10x)
antiGen, val−selected (10x)
Mean of z−scores
Mean of log−probs
Mean of ranks
BA.2 DMS (LY−CoV1404 escape)
BA.2 DMS (expression)
BA.2 DMS (bind)
BA.1 DMS (expression)
BA.1 DMS (bind)
Where × which
Substitution matrix
Tranception
EVE
MSA Trans. (BA.2 eval)
MSA Transfomer
PSSM Conservation
PSSM
Codon table
BLOSUM62
ESM 2 (650M)
ESM 2 (35M)
EVEscape
2nd
3rd
1st
0.00 0.25 0.50 0.75
Max PSSM precision


J
Max precision, de novo
1st
2nd
3rd
0.0 0.2 0.4 0.6
Max PSSM precision

Model type
antiGen
Pandemic Data in Training Window
Ensembles
Deep Mutational Scans
Pre−Pandemic Data Only
Val−selected, averaged over 10 holdouts


K
Recall at q = 0.1, all mutations
antiGen
Where x which + PSSM, val−selected (10x)
antiGen, val−selected (10x)
Mean of z−scores
Mean of log−probs
Mean of ranks
BA.2 DMS (LY−CoV1404 escape)
BA.2 DMS (expression)
BA.2 DMS (bind)
BA.1 DMS (expression)
BA.1 DMS (bind)
Where × which
Substitution matrix
Tranception
EVE
MSA Trans. (BA.2 eval)
MSA Transfomer
PSSM Conservation
PSSM
Codon table
BLOSUM62
ESM 2 (650M)
ESM 2 (35M)
EVEscape
3rd
2nd
1st
0.0 0.2 0.4 0.6 0.8
Recall at q = 0.1


L
Recall at q = 0.1, de novo
3rd
2nd
1st
0.0 0.2 0.4 0.6
Recall at q = 0.1

M
DMS correlation
Corr. (Expression)
0.4
0.2
0.0
antiGen
Mean of log−probs
Mean of ranks
Mean of z−scores
EVE
MSA Trans. (BA.2 eval)
MSA Transfomer
Tranception
BLOSUM62
ESM 2 (35M)
ESM 2 (650M)
EVEscape
antiGen, val−selected (10x)
Where x which + PSSM, val−selected (10x)
tiGen
Ensembles
Pandemic Data in Training Window
Pre−Pandemic Data Only
Val−selecte
Transformation
Mean
Mean, log
None
Log

N
PSSM precision, 100+ de novo
P(mut. in top decile obs. by t)
0.075
0.050
0.025
0.000
antiGen
antiGen as ESM 2 (35M) LM head FT
antiGen as ESM 2 (35M) full FT
antiGen as ESM 2 (35M) partial FT
antiGen as EVE full FT
antiGen as Tranception full FT
2 3 4 5 6
Date (t, years into pandemic)

O
Recall, top decile
P(obs. mut. in top decile)
0.75
0.50
0.25
0.00
antiGen
antiGen / PSSM gate
Site x Substitution
SxS, MCM interpolation
BLOSUM62
ESM−2 35M, partial FT
Where x which
Where x which + events
PSSM
EVEscape
2021 2022 2023 2024
Train / test cutoff